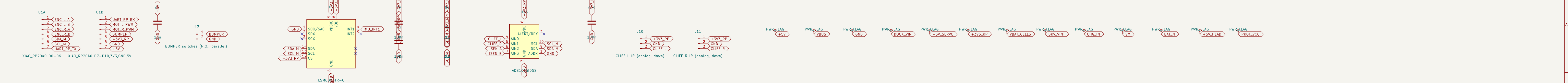
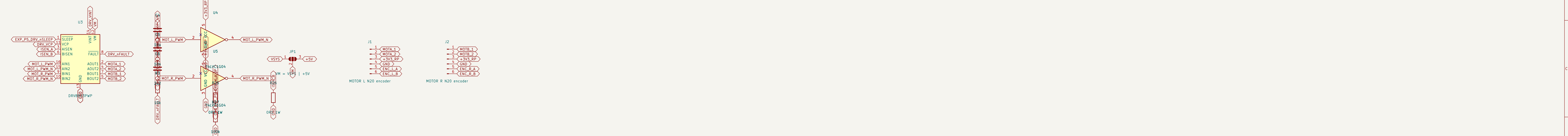


ORBIE V5 – BODY CONTROLLER r0.2 (applies HW requirements 26 Sep 2026). Retrofit outline of the V3 Main PCB drawing – CONFIRM mounting pattern against the body before fab.

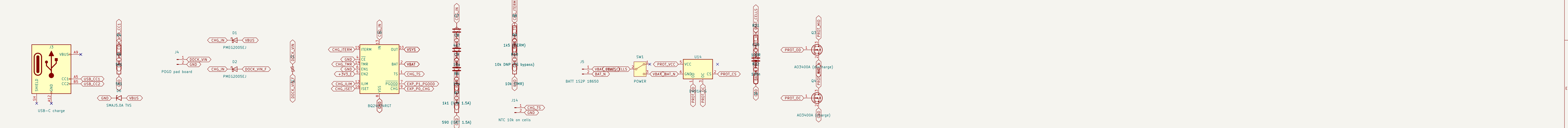
U1 XIAO RP2040 — 1 kHz balance loop. D0–D3 encoders (P0), D4/D5 I2C_M (IMU + ADS1251) D6/D7 I2C_A (AHT10) D8/D9 I2C_B (ADXL345) D10/D11 I2C_C (HTU3D) D12/D13 I2C_D (HTU3D) D14/D15 I2C_E (HTU3D) D16/D17 I2C_F (HTU3D) D18/D19 I2C_G (HTU3D) D20/D21 I2C_H (HTU3D) D22/D23 I2C_I (HTU3D) D24/D25 I2C_J (HTU3D) D26/D27 I2C_K (HTU3D) D28/D29 I2C_L (HTU3D) D30/D31 I2C_M (HTU3D) D32/D33 I2C_N (HTU3D) D34/D35 I2C_O (HTU3D) D36/D37 I2C_P (HTU3D) D38/D39 I2C_Q (HTU3D) D40/D41 I2C_R (HTU3D) D42/D43 I2C_S (HTU3D) D44/D45 I2C_T (HTU3D) D46/D47 I2C_U (HTU3D) D48/D49 I2C_V (HTU3D) D50/D51 I2C_W (HTU3D) D52/D53 I2C_X (HTU3D) D54/D55 I2C_Y (HTU3D) D56/D57 I2C_Z (HTU3D) D58/D59 I2C_AA (HTU3D) D60/D61 I2C_AB (HTU3D) D62/D63 I2C_AC (HTU3D) D64/D65 I2C_AD (HTU3D) D66/D67 I2C_AE (HTU3D) D68/D69 I2C_AF (HTU3D) D70/D71 I2C_AG (HTU3D) D72/D73 I2C_AH (HTU3D) D74/D75 I2C_AI (HTU3D) D76/D77 I2C_AJ (HTU3D) D78/D79 I2C_AK (HTU3D) D80/D81 I2C_AL (HTU3D) D82/D83 I2C_AM (HTU3D) D84/D85 I2C_AN (HTU3D) D86/D87 I2C_AO (HTU3D) D88/D89 I2C_AP (HTU3D) D90/D91 I2C_AQ (HTU3D) D92/D93 I2C_AR (HTU3D) D94/D95 I2C_AS (HTU3D) D96/D97 I2C_AT (HTU3D) D98/D99 I2C_AU (HTU3D) D100/D101 I2C_AV (HTU3D) D102/D103 I2C_AW (HTU3D) D104/D105 I2C_AX (HTU3D) D106/D107 I2C_AY (HTU3D) D108/D109 I2C_AZ (HTU3D) D110/D111 I2C_BA (HTU3D) D112/D113 I2C_BB (HTU3D) D114/D115 I2C_BC (HTU3D) D116/D117 I2C_BD (HTU3D) D118/D119 I2C_BE (HTU3D) D120/D121 I2C_BF (HTU3D) D122/D123 I2C_BG (HTU3D) D124/D125 I2C_BH (HTU3D) D126/D127 I2C_BI (HTU3D) D128/D129 I2C_BJ (HTU3D) D130/D131 I2C_BK (HTU3D) D132/D133 I2C_BL (HTU3D) D134/D135 I2C_BM (HTU3D) D136/D137 I2C_BO (HTU3D) D138/D139 I2C_BP (HTU3D) D140/D141 I2C_BQ (HTU3D) D142/D143 I2C_BR (HTU3D) D144/D145 I2C_BS (HTU3D) D146/D147 I2C_BT (HTU3D) D148/D149 I2C_BU (HTU3D) D150/D151 I2C_BV (HTU3D) D152/D153 I2C_BW (HTU3D) D154/D155 I2C_BX (HTU3D) D156/D157 I2C_BY (HTU3D) D158/D159 I2C_BZ (HTU3D) D160/D161 I2C_CA (HTU3D) D162/D163 I2C_CB (HTU3D) D164/D165 I2C_CC (HTU3D) D166/D167 I2C_CD (HTU3D) D168/D169 I2C_CE (HTU3D) D170/D171 I2C_CF (HTU3D) D172/D173 I2C.CG (HTU3D) D174/D175 I2C_CH (HTU3D) D176/D177 I2C_CI (HTU3D) D178/D179 I2C_CJ (HTU3D) D180/D181 I2C_CK (HTU3D) D182/D183 I2C_CL (HTU3D) D184/D185 I2C_CM (HTU3D) D186/D187 I2C_CO (HTU3D) D188/D189 I2C_CP (HTU3D) D190/D191 I2C_CQ (HTU3D) D192/D193 I2C_CR (HTU3D) D194/D195 I2C_CS (HTU3D) D196/D197 I2C_CT (HTU3D) D198/D199 I2C_CU (HTU3D) D200/D201 I2C_CV (HTU3D) D202/D203 I2C_CW (HTU3D) D204/D205 I2C_CX (HTU3D) D206/D207 I2C_CY (HTU3D) D208/D209 I2C_CZ (HTU3D) D210/D211 I2C_DA (HTU3D) D212/D213 I2C_DB (HTU3D) D214/D215 I2C_DC (HTU3D) D216/D217 I2C_DD (HTU3D) D218/D219 I2C_DE (HTU3D) D220/D221 I2C_DF (HTU3D) D222/D223 I2C.DG (HTU3D) D224/D225 I2C_DH (HTU3D) D226/D227 I2C_DI (HTU3D) D228/D229 I2C_DJ (HTU3D) D230/D231 I2C_DK (HTU3D) D232/D233 I2C_DL (HTU3D) D234/D235 I2C_DM (HTU3D) D236/D237 I2C.DO (HTU3D) D238/D239 I2C_DP (HTU3D) D240/D241 I2C_DQ (HTU3D) D242/D243 I2C_DR (HTU3D) D244/D245 I2C_DS (HTU3D) D246/D247 I2C_DT (HTU3D) D248/D249 I2C_DU (HTU3D) D250/D251 I2C_DV (HTU3D) D252/D253 I2C.DW (HTU3D) D254/D255 I2C_DX (HTU3D) D256/D257 I2C_DY (HTU3D) D258/D259 I2C_DZ (HTU3D) D260/D261 I2C_EA (HTU3D) D262/D263 I2C_EB (HTU3D) D264/D265 I2C_EC (HTU3D) D266/D267 I2C_ED (HTU3D) D268/D269 I2C_EE (HTU3D) D270/D271 I2C_EF (HTU3D) D272/D273 I2C. EG (HTU3D) D274/D275 I2C_EH (HTU3D) D276/D277 I2C_EI (HTU3D) D278/D279 I2C_EJ (HTU3D) D280/D281 I2C_EK (HTU3D) D282/D283 I2C_EL (HTU3D) D284/D285 I2C_EM (HTU3D) D286/D287 I2C.EO (HTU3D) D288/D289 I2C_EP (HTU3D) D290/D291 I2C_EQ (HTU3D) D292/D293 I2C_ER (HTU3D) D294/D295 I2C_ES (HTU3D) D296/D297 I2C_ET (HTU3D) D298/D299 I2C_EU (HTU3D) D300/D301 I2C_EV (HTU3D) D302/D303 I2C.EW (HTU3D) D304/D305 I2C_EX (HTU3D) D306/D307 I2C_EY (HTU3D) D308/D309 I2C_EZ (HTU3D) D310/D311 I2C_FA (HTU3D) D312/D313 I2C_FB (HTU3D) D314/D315 I2C.FC (HTU3D) D316/D317 I2C_FD (HTU3D) D318/D319 I2C_FE (HTU3D) D320/D321 I2C_FF (HTU3D) D322/D323 I2C.FG (HTU3D) D324/D325 I2C_FH (HTU3D) D326/D327 I2C_FI (HTU3D) D328/D329 I2C_FJ (HTU3D) D330/D331 I2C_FK (HTU3D) D332/D333 I2C_FL (HTU3D) D334/D335 I2C_FM (HTU3D) D336/D337 I2C.FO (HTU3D) D338/D339 I2C_FP (HTU3D) D340/D341 I2C_FQ (HTU3D) D342/D343 I2C_FR (HTU3D) D344/D345 I2C_FS (HTU3D) D346/D347 I2C_FT (HTU3D) D348/D349 I2C_FU (HTU3D) D350/D351 I2C_FV (HTU3D) D352/D353 I2C.FW (HTU3D) D354/D355 I2C_FX (HTU3D) D356/D357 I2C_FY (HTU3D) D358/D359 I2C_FZ (HTU3D) D360/D361 I2C_GA (HTU3D) D362/D363 I2C_GB (HTU3D) D364/D365 I2C_GC (HTU3D) D366/D367 I2C.GD (HTU3D) D368/D369 I2C_GE (HTU3D) D370/D371 I2C_GF (HTU3D) D372/D373 I2C.GG (HTU3D) D374/D375 I2C_GH (HTU3D) D376/D377 I2C_GI (HTU3D) D378/D379 I2C_GJ (HTU3D) D380/D381 I2C_GK (HTU3D) D382/D383 I2C_GL (HTU3D) D384/D385 I2C.GM (HTU3D) D386/D387 I2C.GO (HTU3D) D388/D389 I2C_GP (HTU3D) D390/D391 I2C_GQ (HTU3D) D392/D393 I2C_GR (HTU3D) D394/D395 I2C_GS (HTU3D) D396/D397 I2C_GT (HTU3D) D398/D399 I2C_GU (HTU3D) D400/D401 I2C_GV (HTU3D) D402/D403 I2C.GW (HTU3D) D404/D405 I2C_GX (HTU3D) D406/D407 I2C_GY (HTU3D) D408/D409 I2C_GZ (HTU3D) D410/D411 I2C_HA (HTU3D) D412/D413 I2C_HB (HTU3D) D414/D415 I2C_HC (HTU3D) D416/D417 I2C_HD (HTU3D) D418/D419 I2C.HE (HTU3D) D420/D421 I2C_HF (HTU3D) D422/D423 I2C.HG (HTU3D) D424/D425 I2C_HH (HTU3D) D426/D427 I2C_HI (HTU3D) D428/D429 I2C_HJ (HTU3D) D430/D431 I2C_HK (HTU3D) D432/D433 I2C_HL (HTU3D) D434/D435 I2C.HM (HTU3D) D436/D437 I2C.HO (HTU3D) D438/D439 I2C_HP (HTU3D) D440/D441 I2C_HQ (HTU3D) D442/D443 I2C_HR (HTU3D) D444/D445 I2C_HS (HTU3D) D446/D447 I2C_HT (HTU3D) D448/D449 I2C_HU (HTU3D) D450/D451 I2C_HV (HTU3D) D452/D453 I2C.HW (HTU3D) D454/D455 I2C_HX



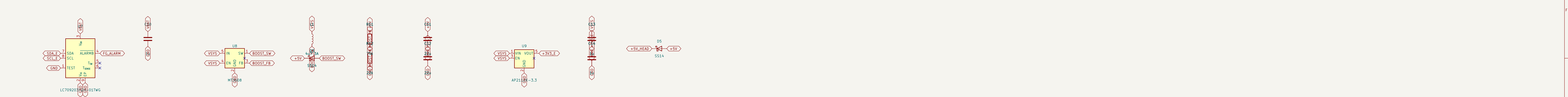
U3 DRV8833 locked-antiphase (IN1 = PWM, IN2 = NOT PWM via U4/U5). nSLEEP from expander, default LOW (pull-down). 0.1R sense -> chopping limit 2 A + ADC current sense



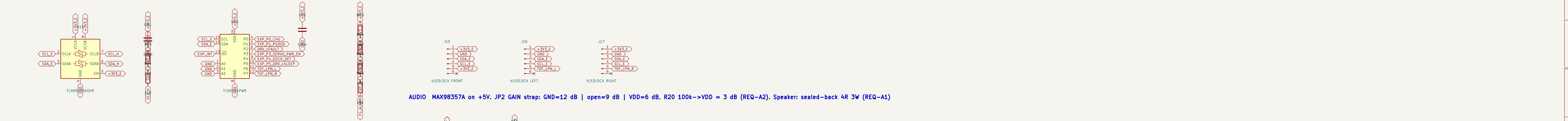
POWER USB-C 6P (TVS, REQ-R5) or belly POGO -> OR diodes -> BQ24074 1S charger/power path -> VSYS. Pack: 1S2P 18650 + DW01A/2xA03400 protection (REQ-P1) + NTC on TS (REQ-P2)



U7 LC709203F fuel gauge @0x08 (battery_pct for REQ-P4/P5) U8 MT3608 5V/2A U9 AP2112K 3V3_E D5 isolates the head 5V so the XIAO's USB can never back-feed a laptop (REQ-R5)



U15 TCA9517A I2C buffer: body bus (A) <--> neck harness (B) so a failing head sensor cannot hang the body (REQ-R1) U10 TCA9534 5V to 1.8V CH3300SD 50AUB 36RVA000W DQ (6) 6666 Tefl R1083 B6P H6666 B6P 1.8V (REQ-N3 option A). Front LPn tied high; L/R LPn from the expander to assign addresses at boot



J7 NECK HARNESS 16p: GND +5V_HEAD 3V3 SDA_H SCL_H UART I2S SERVO IR_RX LIDAR_RX XIAO_RST XIAO_BOOT. J12 LD19 lidar in the base (REQ=N4): 5V, TX->GPIO4 of the head XIAO. SW2/SW3 recessed RESET/BOOT (REQ=X1..X4).

